

5G/LTE Network Performance Summary

Performance Index Definition

Original Formula:

$$\text{Performance} = 0.4 * z(\text{RSRP}) + 0.4 * z(\text{SINR}) + 0.2 * z(\text{Throughput})$$

Components (per RAT):

1. Coverage (RSRP, 40% weight): Signal strength indicator
Normalized: $(\text{RSRP} + 120) / 60$ [dBm range -120 to -60]
2. Quality (SINR, 40% weight): Signal-to-Interference ratio
Normalized: $\text{SINR} / 30$ [dB range 0-30]
3. Capacity (Throughput, 20% weight): Downlink data rate
Normalized: $\min(\text{TP} / 200, 1.0)$ [Mb/s]

Output Range:

0 (poor) to 1 (excellent)

Performance Bands:

1. Low (48825 samples)
Performance < 0.33
2. Medium (45733 samples)
 $0.33 \leq \text{Performance} < 0.67$
3. High (18115 samples)
Performance ≥ 0.67

Key Findings

1. RAT comparison:

NR5G (5G) Avg = 0.418 (N=104807)

LTE Avg = 0.186 (N=9063)

5G outperforms LTE by 23.2%

2. Spatial Coverage:

- Grid Cells: 7097
- Average samples per cell: 16.0
- Geographic bounds
Latitude [42.829, 43.251]
Longitude [-88.555, -87.848]

3. Missing Value Handling:

- Strategy
Mask component if any KPI is missing
Skip record if all missing
- NaN rates
RSRP ~40%
SINR ~40%
Throughput ~40%
- Records with valid Performance
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Limitations & Future Steps

1. Single market (Milwaukee) passed, multi-market comparison required
2. 1-second aggregation skips sub-second dynamics
3. Possible to create interactive Folium maps (Point, Path, Hex layers)